

Praveen Korukonda

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Education

Indian Institute of Technology Roorkee, Uttarakhand, India
Bachelor of Technology, Mechanical Engineering

Oct 2021 – May 2025
GPA: 7.87/10

Relevant Courses

Statistical Machine Learning (MIL-505), Artificial Intelligence (MIN-516), Programming & Data Structures (MIN-103) Mathematics-I (MAN-001), Numerical Methods (MAN-004)

Interests

Large-Scale ML Systems, Applied Machine Learning, Natural Language Processing, Generative AI
Time Series Analysis & Sequence Modeling, Computer Vision, Distributed Systems

Professional Experience

Software Engineering Intern | *Planto AI* Sept 2025 – Dec 2025

- Built an incident management system using microservices architecture with Go and PostgreSQL database
- Designed 15+ REST APIs with unit tests across 4 services to manage end-to-end incident lifecycles
- Reduced Mean Time to Acknowledge (MTTA) by 27% through automated, real-time Slack alerts
- Built an invoice generation service in Go to create dynamic PDF invoices, achieving 55 ms average latency
- Developed secure REST APIs and integrated PostgreSQL for storing and retrieving PDF invoices

Software Engineering Intern | *CodingPro* June 2025 – Aug 2025

- Built an end-to-end secure job portal application using Spring Boot, Thymeleaf and MySQL database
- Containerized the application using Docker and deployed it to AWS Elastic Beanstalk with RDS for storage
- Implemented role-based authentication with Spring Security to secure both candidate and recruiter profiles
- Developed e2e recruitment workflows, profile management, job application tracking, and resume uploads

Research Experience

Bachelor Thesis | Estimation of RUL of Bearings Using Machine Learning Sep 2024 – May 2025

Supervisor: *Prof. S.P. Harsha*

- Built a predictive-maintenance framework estimating Remaining Useful Life of high-speed bearings
- Extracted 34 multi-domain features and filtered them via S-weighted and Pearson correlation scoring
- Designed a self-attention-augmented BiLSTM to model temporal degradation patterns in vibration data
- Tuned hyperparameters via Bayesian optimisation, selecting hidden units, dropout, and learning rate
- Achieved R^2 of 0.825 on combined data and 0.781 on the held-out bearing under leave-one-out testing

Research Internship | Simulation of Human Induced Load Patterns Using ML July 2024 – Aug 2024

Supervisor: *Prof. Anil Kumar*

- Developed a GAN-based model generating human-induced load patterns at 98% similarity to real data
- Processed 658+ experimental samples across walking, jumping, and bouncing loads at 1.5 to 3.5 Hz
- Built a Forecasting+LSTM hybrid model and studied ARIMA and Prophet for time-series load synthesis
- Validated synthetic loads via PSD replication and statistical equivalence in structural response

Projects

Bytshell | *Association for Computing Machinery (ACM), IIT Roorkee* [github/byte-shell](#)

- Built a Unix command line shell from scratch in C that used to execute built-in Linux commands
- Implemented core features like directory navigation and process handling using low-level unix system calls

Image Denoiser | *Vision and Language Group (VLG), IIT Roorkee* [github/image-denoiser](#)

- Designed an Autoencoder Model achieving PSNR of 28.256 for noise reduction in low-light images.
- Applied RIDNet, enhancing image visibility and detail retention with PSNR of 30.579.

Stock Sentiment Analysis Using ML | *Finance Club, IIT Roorkee* [github/stock-sentiment](#)

- Developed a machine learning model predicting stock movements from news headlines using NLP
- Backtested a trading strategy yielding 17.12% returns and a 1.34 Sharpe ratio

Torrent-X | *Personal Project* [github/torrent-x](#)

- Created a BitTorrent client in Java that allows downloads using a single-file torrent or magnet links
- Implemented p2p protocol with tracker communication, piece verification, and metadata exchange

mini-redis | *Personal Project* [github/mini-redis](#)

- Developed an in-memory Redis-style server and a CLI client in Go that supports core Redis commands
- Implemented thread-safe storage for strings (with expiry), lists (with blocking operations), and sorted sets
- Supports multiple concurrent clients and loads an existing RDB file at startup for basic persistence

git-lite | *Personal Project* [github/git-lite](#)

- Created a Git client in Java that clones Github repositories using Git's Smart HTTP, which is simple and fast

caching-proxy | *Personal Project* [github/caching](#)

- Built a caching reverse proxy command-line tool in Go that reduces server hits and request latency
- Developed a thread-safe and concurrent disk-based cache storage system using SHA-256 hashing

go-lb | *Personal Project* [github/go-lb](#)

- Designed a load balancer in Go that supports concurrent request handling and active/passive health checks

Honors & Achievements

- Secured an All-India-Rank of 6026 in JEE Main 2021 amongst 1.5 Million candidates
- Secured an All-India-Rank of 7514 in JEE Advanced 2021 amongst 250,000 candidates
- Awarded Merit-cum-Means (MCM) Scholarship for four consecutive years at IIT Roorkee

Skills

Languages: Python, Golang, Java, C/C++, JavaScript, SQL

Tools & Technologies: TensorFlow, PyTorch, Scikit-learn, Pandas, Git, Docker, AWS, REST APIs, CI/CD, React MySQL, PostgreSQL, Linux, Jupyter, Google Collab, SpringBoot

CS Fundamentals: System Design, Data Structures, Algorithms, Objected Oriented Programming, Operating Systems, Database Management System

University Service

Technical Council Member | *Govind Bhawan Hostel, IIT Roorkee* March 2024 – June 2024

- Collaborated with hostel administration to improve technical infrastructure and quality of internet services
- Organized technical workshops & hackathons to enhance technical skills among 600+ hostel students

Contingent Leader | *Student Technical Council, IIT Roorkee* April 2024

- Led a contingent of 60 students, representing my hostel in the annual Inter Hostel Tech Fest 2024
- Led participation across 8 diverse technical Problem Statements & guided the teams to secure 4 Silver Medals